

Exploring the Design of a Multimodal Head-Mounted Virtual Reality for Earthquake Escape Education for Primary School Students

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Abstract

Earthquakes pose significant risks to the safety and mental well-being of primary school students, yet traditional earthquake-escape education methods, such as lectures and real-life drills, often fail to engage young learners or effectively improve their emergency response skills. To address these limitations, we conducted formative research with primary school students, teachers, and earthquake rescue experts in a seismic zone with a history of severe earthquakes. Based on these insights, we designed a multi-modal, head-mounted virtual reality (VR) experience that simulates earthquakes and trains children in effective escape techniques. Our approach combines multi-modal interactions (notably vibration), in-situ personalized guidance, and gamification to create an immersive and practical learning environment. We discuss key considerations for designing effective multi-modal VR environments to enhance earthquake escape education for children.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interactive systems and tools**;

Keywords

Virtual Reality (VR), Multi-modal Design, Earthquake Escape Education, Children

ACM Reference Format:

Renxuan Ye and Min Fan. 2025. Exploring the Design of a Multimodal Head-Mounted Virtual Reality for Earthquake Escape Education for Primary School Students. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3706599.3719919>

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/2025/04

<https://doi.org/10.1145/3706599.3719919>

1 Introduction

Natural disasters, such as earthquakes, pose significant threats to people's safety and mental health, particularly for those living in cities located in seismic zones. Despite the widespread implementation of earthquake safety education in schools, current methods primarily rely on traditional approaches such as lectures and drills, which often fail to effectively engage young students or equip them with adequate emergency response skills [3]. These methods lack realism and interactivity, limiting students' ability to understand and apply what they learn in emergency scenarios.

Immersive head-mounted Virtual Reality (VR) technology, with its unique features of immersion, interaction, and simulation, has emerged as a promising tool for disaster education [4]. VR, particularly with multi-modal features, enables students to experience realistic emergency scenarios in a safe and controlled environment, fostering both better understanding and skill development. However, existing VR applications for emergency training often focus on fire safety or other types of disaster rescues [5, 8], with limited attention given to earthquake-specific challenges. Unlike other disasters, earthquakes present unique risks, such as unpredictable structural collapses, aftershocks, and ground vibrations [6], which demand specialized training to cultivate rapid and effective responses.

In this paper, we explore the design space of immersive VR experiences to enhance earthquake escape education for children. Our research question investigates how to design an effective multi-modal VR environment that improves children's serious awareness of earthquake disasters, as well as their ability to respond effectively in emergency situations. To address this, we conducted formative research to identify the user needs and core requirements. Based on these findings, we developed a multi-modal VR experience to simulate realistic earthquake scenarios. The system incorporates audiovisual elements and notably seismic vibration simulation, to enhance the realism of the earthquake experiences, while also providing in-situ guidance and feedback for children (Figure 1). We discuss key design considerations for balancing seriousness and gamifications and for integrating in-situ and personalized learning to support knowledge and skill development. Our work contributes to the multi-modal design of serious disaster education tools for children, offering insights into how VR systems can enhance engagement, realism, and practical skill development.

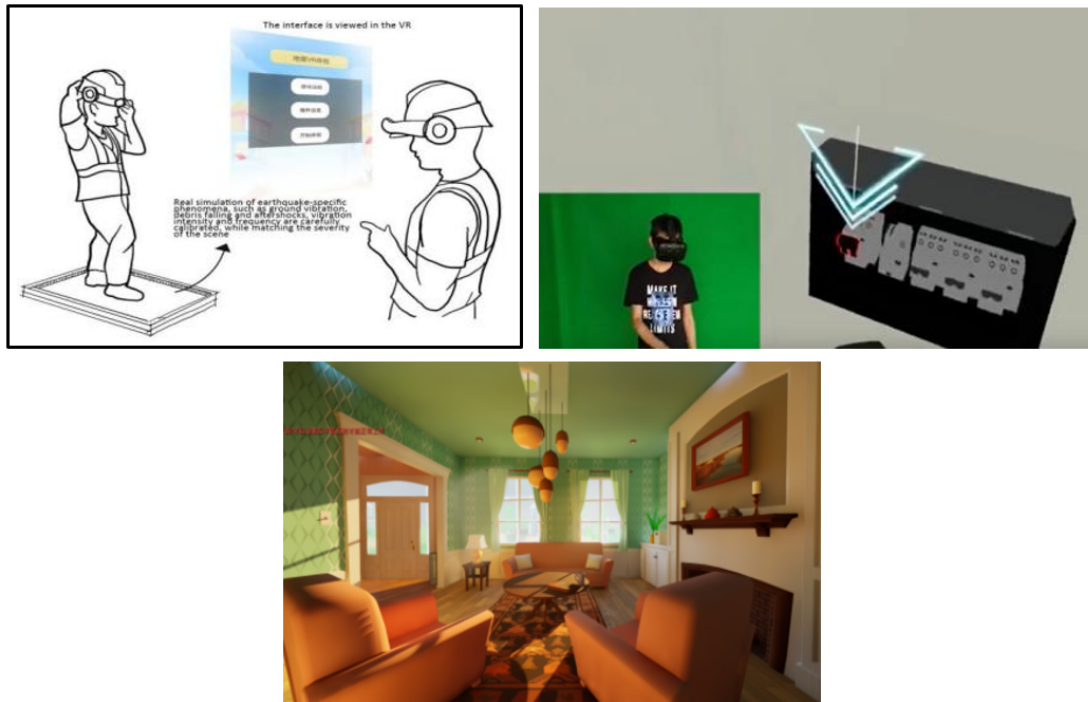


Figure 1: Initial interface and the application scenarios

2 RELATED WORK

2.1 Earthquake Safety Education for Primary School Students

Earthquakes are unique among natural disasters due to their sudden onset, short duration, and the unpredictability of aftershocks, making them particularly challenging to prepare for and respond to [7]. These events often lead to widespread structural collapses, falling debris, and unstable ground conditions, requiring quick, informed decision-making to ensure safety. Unlike disasters such as fires or floods, which typically offer some warning or follow more predictable patterns, earthquakes demand tailored education approaches that emphasize situational awareness, rapid response, and resilience [5].

For primary school students, earthquake safety education is typically delivered through classroom lectures and basic evacuation drills [3]. While these traditional methods provide foundational knowledge, they often lack realism, engagement, and interactivity, which are crucial for equipping young learners with practical emergency response skills [4]. In recent years, digital tools have emerged as valuable supplements to traditional methods. These tools include mobile applications, serious games, and interactive simulations that aim to enhance learning outcomes. For example, Feng et al. (2021) developed a VR-based family earthquake escape game that significantly improved participants' preparedness and survival skills [8]. Similarly, Katsanos et al. (2023) demonstrated the effectiveness of gamified VR simulations in enhancing situational awareness and decision-making in earthquake scenarios.

Despite these advancements, VR-based disaster education solutions specifically tailored for children remain scarce [1].

This gap is particularly concerning given that young learners have unique cognitive and emotional needs, requiring educational approaches that go beyond knowledge delivery to also foster confidence and practical skills in a safe and engaging environment. This paper explores how VR technology, integrated with multi-modal interactions such as visual, auditory, and haptic feedback, can address these challenges. By simulating key earthquake-specific phenomena such as ground shaking, falling debris, and aftershocks, this study seeks to improve the effectiveness of disaster preparedness education for children, bridging the gap between theoretical learning and real-world emergency response.

2.2 Virtual Reality in Emergency Education

Virtual Reality (VR) technology has emerged as a transformative tool in the emergency education sector, offering unparalleled advantages in immersion, interactivity, and experiential learning. By simulating realistic environments, VR allows learners to engage in hands-on training without the risks associated with real-life scenarios. This is particularly beneficial in emergency education, where students can practice responding to high-stress situations in a safe, controlled environment. For instance, Farra et al. (2013) demonstrated that VR-based disaster training not only enhances participants' emergency response skills but also improves their psychological readiness, a critical factor in effectively managing crises [4].

VR applications have expanded to target specific types of emergencies, such as fires, floods, and earthquakes. Yang et al. (2023)

developed an immersive VR earthquake escape system designed for primary school students, which significantly improved their disaster awareness, survival skills, and confidence in handling earthquake scenarios [5]. These studies underscore the unique capabilities of VR in disaster education, particularly its ability to engage learners through interactive, scenario-based training.

Despite these advancements, the majority of VR-based disaster education research and applications have focused on adult training. Studies specifically tailored to younger audiences, such as primary school students, remain limited. This is particularly concerning in the context of earthquake safety education, where young learners often face heightened vulnerability due to their limited cognitive and emotional development. While some VR tools for children have been introduced, they often fail to address the unique challenges posed by earthquakes, such as sudden ground shaking, rapid environmental changes, and unpredictable aftershocks.

2.3 Multimodal Interaction and Gamification in VR Education

Dual coding theory emphasizes the effectiveness of combining visual and auditory channels for optimized learning outcomes. Haptic feedback enhances immersion in VR environments by providing realistic physical sensations, improving response accuracy and confidence. Studies such as Feng et al. (2021) have shown that integrating haptic interactions in earthquake training games significantly enhances situational awareness and evacuation skills. Similarly, Katsanos et al. (2023) demonstrated the benefits of tactile cues in fire evacuation VR systems, where sensations like heat and surface changes helped users better navigate hazards [1]. These examples highlight the value of haptics in multimodal VR for disaster education and skill development. Studies have shown that tactile feedback, such as seismic vibration simulation, enables learners to physically experience disaster scenarios, enhancing their situational awareness and decision-making capabilities [1, 8].

Game-based elements (gamifications) such as challenges, rewards, and narrative-driven scenarios are also often used in immersive VR to make learning engaging and impactful [2]. Research indicates that VR with gamifications not only improve knowledge retention but also foster critical thinking and practical skills through interactive and realistic scenarios. For example, Katsanos et al. (2021) demonstrated how immersive VR earthquake games can observe evacuation behaviors and improve preparedness [1]. Similarly, Feng et al. (2021) highlighted the effectiveness of integrating game mechanics into VR disaster training for children, significantly improving their skills and confidence in emergency response [8]. However, the integration of multimodal design and gamifications in VR-based earthquake education for primary school students remains underexplored.

3 Formative Research

We systematically conducted formative research to extract key design requirements for earthquake escape education (Figure 2). Drawing from prior studies and involving children, teachers, and disaster experts in seismic zones, we organized our findings into three high-level design elements: multi-modal interaction, immersion, and personalization. These elements, supported by related

literature and expert input, were further refined into middle-level concepts to guide the development of the virtual reality-based system.

3.1 Study Methods

This study used a combination of qualitative and quantitative research methods to explore the requirements and challenges of earthquake escape education for primary school students. Semi-structured interviews were conducted with 10 primary school teachers and 5 disaster management experts to gather insights on educational objectives, limitations of current methods, and recommendations for improvement. Additionally, a survey involving 100 primary school students was carried out to assess their existing knowledge, learning preferences, and engagement levels with disaster education. Classroom observations of three earthquake drills were also performed to analyze students' practical responses and the effectiveness of traditional training. The collected data were analyzed thematically and statistically, providing a basis for the design of a virtual reality-based multi-modal system. These findings particularly informed the inclusion of earthquake vibration simulation equipment to enhance tactile realism, aiming to address the shortcomings of traditional educational methods.

3.2 Primary Results

The primary results of this study were derived from interviews with educators, disaster management experts, and observations of current earthquake education practices. These findings highlight key educational objectives for primary school students in earthquake escape training: not only to impart emergency knowledge and skills but also to enhance their overall safety awareness. Feedback from experts emphasized the limitations of traditional methods, such as low engagement and the lack of realistic scenarios for practical skill development. Based on these insights, the study identified the need for a multi-modal approach that integrates visual, auditory, and haptic elements to create an immersive and effective learning environment.

For instance, experts suggested incorporating realistic tactile feedback to simulate varying seismic intensities, which could improve students' situational awareness and decision-making under stress. Additionally, educators highlighted the importance of gamified elements—such as challenges and rewards—to maintain engagement, as well as personalized guidance to address the diverse needs of learners. These results informed the subsequent system design, aiming to address the shortcomings of traditional earthquake education and better align with the practical requirements of disaster preparedness training.

4 SYSTEM DESIGN AND IMPLEMENTATION

4.1 Learning and Design Goals

The systematic educational objectives include (1) improving students' disaster knowledge reserve and emergency awareness, (2) cultivating rapid response ability through virtual multi-modal training, and (3) enhancing learning interest of gamified experiences.

The design goals include: (1) providing a multi-modal learning experience that integrates audiovisual, kinetic, and haptic feedback for enhanced immersion and realism; (2) enabling dynamic system

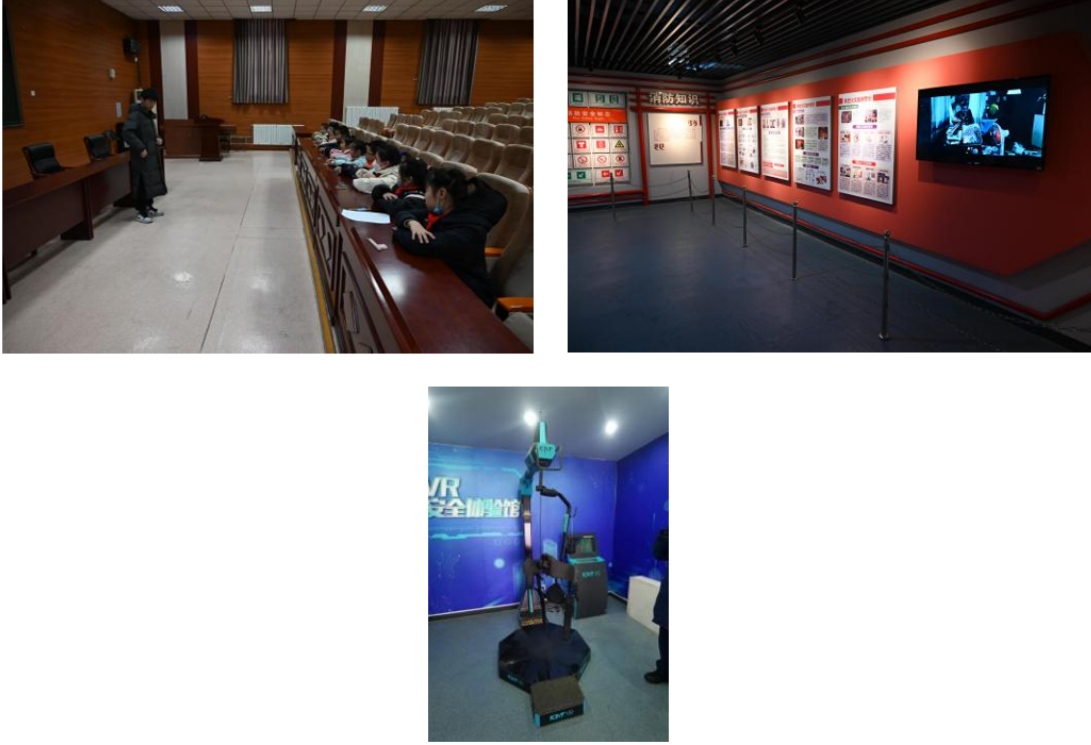


Figure 2: We conducted field studies with the expert team of the Emergency Management Bureau, the teachers and students at the affected primary schools, the memorial halls and the fire brigades, etc., to investigate the current immersive application scenarios.

adaptation with personalized guidance and feedback based on user performance; and (3) incorporating gamification to motivate learners through task-driven challenges and reward mechanisms while maintaining educational focus.

4.2 System Overview

The system is composed of advanced hardware, including a high-definition VR headset, seismic vibration simulation equipment, and motion tracking sensors. These components provide high-quality visual output, realistic tactile feedback replicating earthquake vibrations, and real-time interaction tracking. Together, they create an immersive and responsive learning environment for earthquake escape education.

In this system, users take on the role of a student tasked with escaping from a simulated environment, such as a home or school building, within a set time limit. They navigate hazards like falling debris and structural collapses by assessing risks and identifying safe zones. Through both free exploration and guided task modes, users engage in realistic scenarios that enhance their situational awareness, critical thinking, and decision-making skills, ensuring a comprehensive and engaging educational experience.

The system provides two interactive modes: free exploration and task guidance. In the free exploration mode, students can independently evaluate the seismic environment, identify potential dangers, and practice avoiding hazards. In the task guidance mode, students'

complete goal-oriented training tasks with prompts provided by the system, enabling structured learning. The data recording module captures students' behavior trajectories and decision-making performance in real time, generating personalized learning reports at the end of each session. These reports provide effective feedback for students and teachers, helping to enhance the precision and effectiveness of follow-up education and training.

4.3 Key Design Features and Implementation

4.3.1 Multimodal Interaction. The multi-modal interaction seamlessly integrates motion sensors, gesture recognition, and feedback synchronization to deliver a highly immersive earthquake simulation experience. Powered by Unreal Engine (UE), it generates realistic seismic scenarios, including falling debris, structural collapses, and environmental hazards, presenting unpredictable challenges that enhance situational awareness and decision-making skills. High-resolution VR environments provide dynamic visual feedback, while directional audio offers spatial cues, such as alarms and the sound of collapsing structures. Tactile feedback, delivered through seismic vibration simulation equipment, authentically replicates earthquake sensations, ranging from subtle tremors to intense shaking. This cohesive system bridges theoretical knowledge with hands-on practical training, offering an intuitive and realistic emergency response learning experience.

Haptic feedback is one of the key features of the system. Through earthquake vibration simulation equipment, the system reproduces realistic seismic vibrations, ranging from light tremors to strong shaking, offering users an immersive sensory experience. The system is fully synchronized with visual and auditory feedback, such as falling debris or collapsing walls, which trigger corresponding vibrations and sound effects to enhance situational perception. To ensure smooth performance, the system adopts advanced optimization strategies, such as Level of Detail (LOD) rendering, which reduces the computational load by simplifying distant objects while retaining high-quality detail for close ones. Additionally, asynchronous asset loading ensures seamless scene transitions without delays, even in highly dynamic scenarios.

4.3.2 In-Situ Scenario Adaptation. The system supports real-time scenario adaptation by analyzing user behaviors, such as explored locations and decision-making patterns, and dynamically adjusting the earthquake simulation. For example, when users enter dangerous areas, the system not only introduces additional challenges, like falling debris or intensified vibrations, but also provides personalized guidance, such as visual cues highlighting safe zones or verbal instructions on evacuation strategies. If a user struggles, the system delivers tailored, step-by-step support to reinforce key safety concepts. This dynamic and individualized approach ensures users engage deeply with realistic scenarios while acquiring practical skills and knowledge effectively.

Unreal Engine (UE)-based physical modelling techniques drive environmental elements in a scene, such as tumbling furniture, broken walls and raised dust, details generated simultaneously with particle systems and dynamic lighting, reinforcing visual realism. Directional audio provides users with spatial awareness, such as simulating the sound of a building collapsing from a specific direction.

4.3.3 Gamified Learning Experience. The system leverages gamified design to enhance student engagement and learning outcomes by combining multi-modal interactions, narrative-driven scenarios, and task-reward mechanisms. The multi-modal interaction design integrates visual, auditory, and tactile feedback to create a highly immersive training experience. Seismic vibration simulation equipment serves as a core innovation, introducing haptic feedback to address the limitations of traditional VR systems. Visual feedback uses high-resolution 3D modelling to vividly recreate seismic scenarios, including collapsing structures and shifting furniture, helping students intuitively understand disaster risks. Auditory feedback provides spatial cues, such as alarms and collapsing walls, through directional audio, while tactile feedback simulates vibration intensity and frequency changes, replicating the physical sensations of earthquakes and enhancing the realism of the training (Figure 3).

Gamified elements, such as task-driven challenges and narrative-based scenarios, immerse students in simulated earthquake situations. Tasks include identifying safe shelters, avoiding hazards, and completing evacuation routes under dynamic conditions like aftershocks. A task-reward system tracks progress and performance, rewarding students for timely and effective responses, fostering a sense of achievement and motivation. This design enhances critical thinking, decision-making, and problem-solving skills, providing an

engaging and effective learning experience that prepares students for real-world emergencies.

5 Discussion, Conclusion, and Future work

This study presents a multi-modal VR system for earthquake escape education tailored to the cognitive and emotional needs of students. We discuss key considerations for designing multi-modal VR for disaster educations for primary school children.

First, by integrating visual, auditory, and haptic feedback, the system creates a highly immersive learning environment that realistically simulates earthquake-specific phenomena such as ground shaking, falling debris, and aftershocks. Unlike traditional methods or non-haptic VR systems [1], the incorporation of haptic feedback enhances realism and provides tactile cues that help children associate sensations with danger, reinforcing key rescue knowledge. The vibration intensity and frequency are carefully calibrated to balance realism with user comfort, avoiding fatigue or discomfort. While the current system provides adjustable seismic feedback to match scenario severity, further research is needed to refine these parameters and explore adaptive designs that dynamically respond to individual users' needs, enhancing engagement and long-term learning outcomes. A comparison between camera shake and physical vibration effects could also be valuable, as camera shake may enhance visual immersion by simulating spatial dynamics, while physical vibration offers a tactile understanding of intensity and direction, with their combination potentially providing a more comprehensive and impactful experience.

Second, the in-situ feedback, personalized guidance, and performance evaluation enhance the system's practical relevance by tailoring each training session to the user's individual needs and performance. Personalized guidance ensures dynamic and varied experiences, such as adjusting evacuation routes or providing context-specific instructions, which motivates users to engage in repeated training. Additionally, it allows the system to observe individual behaviors and target specific areas for improvement. However, since the guidance relies on predefined rules, its richness and adaptability are somewhat limited. Integrating artificial intelligence (AI) could address this by enabling real-time, context-aware guidance and more diverse challenges, though this would increase system complexity and development demands.

Third, the system's task-based design, combined with gamification mechanics, strikes a critical balance between engagement and educational focus. Gamified scenarios, such as identifying safe zones and navigating escape routes, motivate students and foster deeper learning, aligning with findings by Feng et al. (2021) and Katsanos et al. (2023) on the effectiveness of serious games in disaster training [1, 8]. However, this system carefully avoids excessive gamification, which could undermine the seriousness of disaster preparedness.

In conclusion, this work explores the potential for designing multi-modal VR environments for disaster education, particularly for younger audiences. We conducted formative research to identify the design requirements, and we designed a multi-modal, head-mounted VR experience that simulates earthquakes and trains children in effective escape techniques. We discuss the considerations for including multi-modal interactions (notably vibration), in-situ

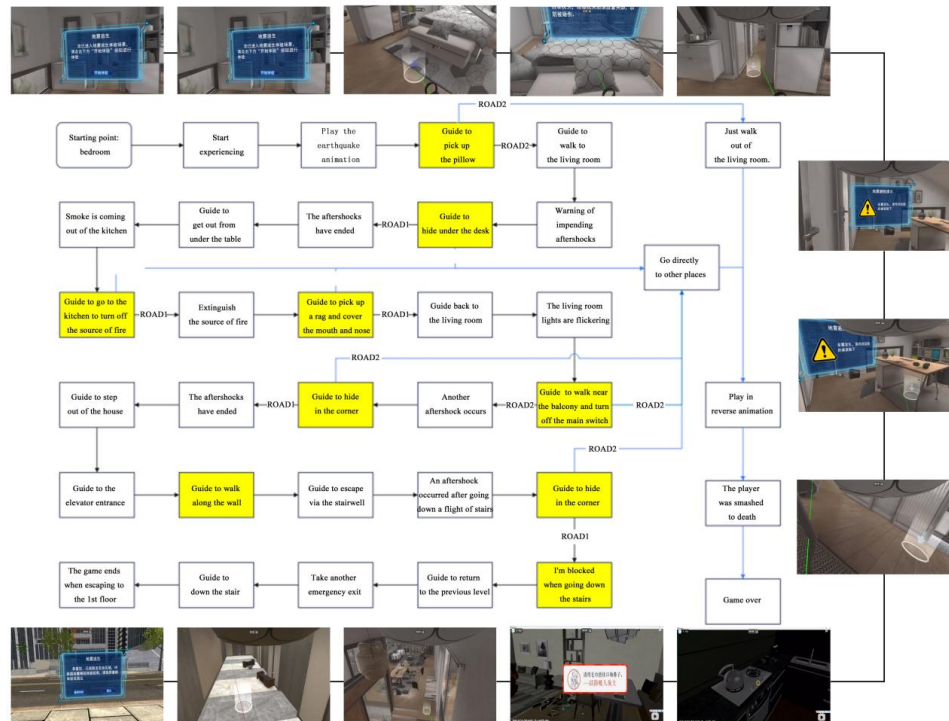


Figure 3: System flowchart and the interaction logic of its escape level. The home earthquake escape level is set for the following three: the first level is the living room, the second level is the kitchen, the third level is the power supply.

personalized guidance, and gamification in creating immersive and practical VR learning environment. Future research includes to conduct a user study with primary school children to further evaluate its system usability and then perform an experiment to.

Acknowledgments

This work was supported by National Social Science Fund of China under grant [22BG137].

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